

PROFESSIONAL DOCUMENTATION

Breakout Swing Scanner Pro Whitepaper

Methodology, architecture, signal logic, repaint controls, and operational limitations.

Script: Breakout Swing Scanner Pro (Daily + Intraday ORB)

Version: 2026.04.24

Prepared: April 24, 2026

Use case: TradingView Pine Script® v6 indicator for chart, watchlist, and Pine Screener workflows.

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Abstract

Breakout Swing Scanner Pro is a TradingView Pine Script® v6 indicator designed to identify breakout candidates across daily and intraday workflows. It combines Donchian-channel breakout logic, Opening Range Breakout structure, volume participation, trend alignment, relative strength, optional market-regime filtering, and ATR-based risk context. The design prioritizes screener compatibility, confirmed-bar operation, and stable higher-timeframe data handling.

1. Problem Statement

Breakout traders often need to scan a large universe quickly without sacrificing interpretability. A useful scanner must expose clean numerical outputs for filtering and sorting, avoid obvious repaint traps, and provide enough context to distinguish a meaningful expansion from an isolated price spike. The original script addressed this need by combining daily and intraday breakout logic. The Pro version formalizes the architecture and adds additional controls for production-style use.

2. Design Objectives

- **Screener-first output:** preserve the first ten `plot()` calls as high-value columns.
- **Confirmed-bar logic:** only fire signals after chart bars confirm.
- **Stable daily context:** use last-confirmed daily values for intraday Donchian, trend, ATR, and benchmark regime context.
- **Configurable selectivity:** allow users to enable or disable volume, trend, VWAP, ADX, relative strength, market regime, and risk filters.
- **Operational clarity:** include dashboard, diagnostic plots, and transparent alert output.

3. System Architecture

Layer	Data source	Role
Chart timeframe	Current symbol OHLCV	Breakout detection, ADX, same-timeframe relative strength, confirmed-bar validation.
Daily symbol context	<code>request.security(syminfo.tickerid, "1D", ...)</code>	Prior daily Donchian levels, prior daily EMA trend context, prior daily ATR for intraday risk.
Benchmark context	User-configurable symbol string, default <code>AMEX:SPY</code>	Relative strength and market-regime gating.
Screener output	First ten <code>plot()</code> calls	Filterable and sortable watchlist columns.
Alert layer	<code>alertcondition()</code> and optional <code>alert()</code>	Watchlist alerts, manual alerts, and webhook-style workflows.

4. Breakout Methodology

4.1 Daily Donchian Breakouts

The daily mode defines long and short levels using prior Donchian extremes:

```
long level = highest(high, Donchian length)[1]
short level = lowest(low, Donchian length)[1]
```

The `[1]` offset excludes the current bar from the level calculation. A long breakout is detected when the selected source crosses above or remains above the long level, depending on the `freshOnly` setting. A short breakdown uses the inverse logic.

4.2 Intraday Opening Range Breakouts

The ORB mode tracks the high and low established during the user-defined opening range. The Pro version calculates the required number of chart bars using a ceiling function so the range is not shortened when the chart interval does not divide the ORB window exactly.

```
orbBars = ceil(openingRangeMinutes / chartBarMinutes)
```

Signals can be generated from the ORB levels, prior daily Donchian levels, or a combined confirmation mode that requires the breakout level to clear the more conservative of the two long/short thresholds.

5. Filter Stack

Filter	Long condition	Short condition	Purpose
Volume	Volume ratio $\geq$ multiplier	Volume ratio $\geq$ multiplier	Requires participation.
EMA trend	Price and fast EMA above slow EMA	Price and fast EMA below slow EMA	Aligns signals with trend direction.
VWAP	Close above session VWAP	Close below session VWAP	Intraday location filter.
ADX	ADX $\geq$ threshold	ADX $\geq$ threshold	Optional trend-strength filter.
Relative strength	Symbol/benchmark ratio above its SMA	Symbol/benchmark ratio below its SMA	Ranks symbol versus benchmark behavior.
Market regime	Prior daily benchmark close above EMA	Prior daily benchmark close below EMA	Reduces counter-regime exposure.
ATR risk	ATR stop distance $\leq$ max %	ATR stop distance $\leq$ max %	Rejects overly wide volatility setups.

6. Repainting and Data Integrity Controls

Two design choices reduce common discrepancies between historical and realtime behavior. First, signals require `barstate.isconfirmed`, so they are evaluated after the current chart bar closes. Second, intraday daily-context requests use prior daily expressions with `lookahead=barmerge.lookahead_on`. This structure returns confirmed higher-timeframe values at the beginning of the next higher-timeframe interval and avoids relying on a daily bar while it is still forming.

```
[dDonHigh, dDonLow, dEmaFastPrev, dEmaSlowPrev, dAtrPrev] = request.security(
    syminfo.tickerid,
    "1D",
    [ta.highest(high, donLen)[1], ta.lowest(low, donLen)[1],
     ta.ema(close, emaFastLen)[1], ta.ema(close, emaSlowLen)[1], ta.atr(atrLen)[1]],
    lookahead = barmerge.lookahead_on)
```

The script is designed to reduce repaint risk, not to eliminate every possible difference caused by data corrections, exchange feed changes, symbol session settings, or user chart configuration.

7. Screener Compatibility

The Pine Screener requires usable `plot()` or `alertcondition()` outputs and exposes plots as columns. The Pro version deliberately keeps the first ten plots focused on actionable scan fields: long signal, short signal, long level, short level, volume ratio, ADX, VWAP deviation, relative-strength ratio, suggested stop price, and mode code.

Screener constraint	Design response
Only plots can be table columns.	All scanner data is emitted through plots.
First ten plots are automatically added.	First ten plots are reserved and ordered by importance.
<code>input.symbol()</code> is ignored by Pine Screener.	Benchmark uses <code>input.string()</code> .
Limited <code>request.*()</code> calls in the scanned script.	The script uses three core request calls, below the documented limit.
Custom timeframe inputs are not supported.	The script relies on the chart/Screener timeframe rather than <code>input.timeframe()</code> .

8. Risk Model

The ATR stop is not an order and is not a complete risk-management system. It is a volatility reference intended to help users compare setup distance across symbols. The stop-distance percentage is calculated as:

$$\text{ATR stop risk \%} = (\text{ATR} \times \text{ATR multiplier} / \text{close}) \times 100$$

Because ATR expands during volatile conditions, this field can help identify signals that are technically valid but impractical for a trader's account size, liquidity constraints, or maximum loss threshold.

9. Deployment Workflow

1. **Define the universe:** use liquid watchlists that match the strategy's asset class.
2. **Choose mode and timeframe:** 1D for swing breakouts; 5/15/30 minute intervals for ORB scans.
3. **Start broad:** disable optional filters to estimate base signal frequency.
4. **Add filters gradually:** trend, volume, VWAP, RS, market regime, and ATR risk should be tested independently.
5. **Validate manually:** review chart structure, news, earnings, liquidity, spreads, and higher-timeframe context before considering an entry.
6. **Record outcomes:** track false breakouts, continuation, stop distance, and post-signal volatility by market regime.

10. Limitations

- The script does not backtest, optimize, or place trades.

- Pine Screener calculations are constrained by TradingView's current Screener design and data-window behavior.
- Volume-based filters may be unreliable on instruments with sparse, synthetic, or unavailable volume.
- ORB logic depends on the chart interval and session definition; mismatched bar times can change the practical opening range.
- Relative strength and market-regime logic use a benchmark symbol; inappropriate benchmarks can produce misleading filters.
- Confirmed-bar operation intentionally trades speed for stability. A signal may appear later than an intrabar price touch.
- Breakout systems can underperform in mean-reverting or news-driven environments.

11. Version Notes

Version	Date	Notes
Original	User-supplied draft	Daily + intraday breakout scanner with screener columns.
Pro 2026.04.24	April 24, 2026	Improved HTF handling, session behavior, ORB sizing, benchmark configurability, risk filter, dynamic alerts, diagnostics, and documentation.

12. Governance and Compliance

Use the indicator as a decision-support tool, not a standalone trading system. Before using it in a live workflow, test it on historical data, compare symbol groups, review alert behavior, and document rules for position sizing, risk limits, and trade invalidation. Users operating in regulated environments should confirm that signal usage, client communications, and record retention comply with all applicable rules.

References

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